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Effects of crossing distance on performance of the native wildflower *Lobelia siphilitica*: Implications for ecological restoration[1]

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CARUSO, C. M., A. M. BENSCOTER, N. V. GALE, E. K. SEIFERT, E. R. MILLS (Department of Integrative Biology, University of Guelph, Guelph, Ontario N1G 2W1 CA), AND A. L. CASE (Department of Biological Sciences, Kent State University, Kent, Ohio 44242 U.S.). J. Torrey Bot. Soc. 142(2): 140–151. 2015—Outbreeding depression, a decline in offspring fitness resulting from mating between distantly related individuals, can jeopardize the success of habitat restoration projects. Concern about outbreeding depression has led to the recommendation that restoration projects use seed collected from geographically proximate populations. However, this recommendation assumes that (1) geographically proximate populations are more ecologically similar than distant populations, and (2) offspring of crosses between individuals from ecologically similar populations have higher fitness than offspring of crosses between ecologically dissimilar populations. We tested these assumptions in *Lobelia siphilitica*, a species commonly used in ecological restoration. We grew F_1 offspring of within- and between-population crosses in the greenhouse and measured four components of performance: seeds per fruit, seed size, germination success, and final aboveground biomass. We found mixed support for the recommendation that restoration projects use seed collected from geographically proximate populations because of concerns about outbreeding depression. Geographically proximate *L. siphilitica* populations had more similar annual mean temperatures than distant populations, as expected if geographically proximate populations are more ecologically similar. However, there was no relationship between geographic distance and the difference in annual precipitation. We also found that crosses with climatically dissimilar populations decreased the performance of *L. siphilitica*, as expected if there is outbreeding depression. However, we detected outbreeding depression in only a subset of populations. Consequently, our results support the recommendation that factors other than geographic distance, particularly climatic distance, need to be considered when selecting seed sources for ecological restoration projects.

Key words: heterosis, hybrid fitness, outbreeding depression, restoration ecology, seed transfer zones.

Mating between individuals from distant populations can either increase or decrease the fitness of their offspring (Lynch 1991). If mating between individuals from distant populations masks deleterious recessive alleles or creates favorable epistatic interactions, then their offspring can have higher fitness relative to the offspring of crosses between geographically proximate populations (*i.e.*, heterosis; Hufford and Mazer 2003). Alternatively, if mating between individuals from distant populations disrupts co-adapted gene complexes or introduces maladaptive alleles to locally adapted populations, then their offspring can have lower fitness relative to the offspring of crosses between geographically proximate populations (*i.e.*, outbreeding depression; Waser and Williams 2001, Edmands 2007). Although the genetic mechanisms that can lead to heterosis and outbreeding depression are well-understood (Lynch and Walsh 1998), it has been difficult to predict the fitness effects of crosses between populations (Edmands and Timmerman 2003), and crossing studies have found evidence for both heterosis (*e.g.*, Pickup et al. 2013) and outbreeding depression (*e.g.*, Hufford et al. 2012).

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Whether mating between individuals from distant populations results in heterosis or outbreeding depression has implications for the success of habitat restoration projects (Hufford and Mazer 2003, McKay et al. 2005). If there is outbreeding depression, then offspring of crosses between restored populations and native populations could have low fitness, jeopardizing both the success of the restoration project and the persistence of the native populations (Montalvo and Ellstrand 2001). Concern about outbreeding depression, among other issues, has led to the recommendation that habitat restoration projects use seed collected from geographically proximate populations (McKay et al. 2005). However, this recommendation is based on at least two assumptions (McKay et al. 2005, Broadhurst et al. 2008). First, geographically proximate populations are assumed to be more ecologically similar than distant populations, and thus likely to be adapted to the same ecological conditions. Second, offspring of crosses between individuals from ecologically similar populations are assumed to have higher fitness than offspring of crosses between ecologically dissimilar populations. If these assumptions do not hold, then the recommendation that habitat restoration projects use seeds from geographically proximate populations might need to be modified (Broadhurst et al. 2008).

We tested these two assumptions in the native North American wildflower *Lobelia siphilitica* L, which is commonly used in prairie (Molano-Flores 2004), woodland (Mottl et al. 2006), and wetland (Bohnen and Galatowitsch 2005) restorations. Although some of these restorations have used seed collected from geographically proximate populations (*e.g.* Bohnen and Galatowitsch 2005), it is not known whether there is outbreeding depression in *L. siphilitica.* We tested for outbreeding depression in *L. siphilitica* by measuring the performance of F_1 crosses both within populations and between populations that were separated by <1 km to 984 km. To test whether geographically proximate populations are more ecologically similar than distant populations, we determined whether the geographic distance between *L. siphilitica* populations was correlated with their climatic distance, measured as the between-population difference in annual mean temperature and annual precipitation. If geographic and climatic distance are correlated, then crosses between geographically distant populations should have lower performance than crosses between nearby populations. To test whether the offspring of crosses between individuals from ecologically similar populations have higher fitness than offspring of crosses between ecologically dissimilar populations, we measured the effect of the climatic distance between *L. siphilitica* populations on the performance of their offspring. Finally, we tested whether crosses between *L. siphilitica* populations have lower performance than crosses within populations, as expected if there is outbreeding depression.

Materials and Methods. Study Species. *Lobelia siphilitica* is a short-lived, herbaceous perennial wildflower that grows in wet prairies and woods throughout eastern North America (Johnston 1991a and references therein, C. M. Caruso, personal observation). Plants produce either female or hermaphroditic flowers (*i.e.*, gynodioecy; Beaudoin Yetter 1989), and female frequency varies widely among populations (0–100%; Caruso and Case 2007). Plants flower from early August until early October and are primarily pollinated by Bombus *spp.* (Beaudoin Yetter 1989). Fruits dehisce from mid-September through November and contain numerous seeds that are passively dispersed (C. M. Caruso and A. L. Case, personal observation). *Lobelia siphilitica* seeds will germinate after 8 weeks of stratification (Johnston 1992), suggesting that they exhibit physiological dormancy (Baskin and Baskin 2004). Perfect flowers are protandrous and pollen is shed from a tube formed by the fused anthers and filaments (Johnston 1991a). The opportunity for autonomous self-fertilization is low because the staminate and pistillate phases of development do not overlap (Johnston 1991b). *Lobelia siphilitica* can also reproduce clonally via rosettes that overwinter and produce a flowering stalk the following summer (Beaudoin Yetter 1989).

Germination and Growth. As part of a larger project to test models of sex ratio evolution in gynodioecious plants (Case and Caruso 2010, Caruso and Case 2013), we germinated seeds from 26 *L. siphilitica* populations in Illinois, Indiana, Iowa, Ohio, and Ontario (Table 1). Seeds were germinated in two separate experiments because of space limitations, and the methods used to break

Table 1. *Lobelia siphilitica* populations included in this study.

Population	State or province	Location	Home or away?	Year field-collected seeds germinated
Bath II	OH	41.181″ N, 81.653″ W	Away	2009
Bath V	OH	41.185″ N, 81.646″ W	Away	2009
Bend View II (BVII)	OH	41.449″ N, 83.788″ W	Home and Away	2009 and 2011
Campbell Prairie	OH	41.535″ N, 83.839″ W	Away	2009
Camp Dodge I	IA	41.688″ N, 93.720″ W	Away	2011
Carthage (CAR)	IL	40.313″ N, 91.042″ W	Home	2009 and 2011
CERA	IA	41.682″ N, 92.870″ W	Away	2011
Crawford Lake	ON	43.465″ N, 79.947″ W	Away	2009
Harker's Run (HR)	OH	39.508″ N, 84.716″ W	Home	2009 and 2011
Jennings Woods Creek	OH	41.172″ N, 81.202″ W	Away	2009
Kaylor Road (KYR)	OH	40.607″ N, 81.599″ W	Home and Away	2009
Krumm	IA	41.705″ N, 92.786″ W	Away	2011
Lions Club	ON	43.219″ N, 79.999″ W	Away	2009
Martin Road	ON	43.226″ N, 80.007″ W	Away	2009
McCormick Bridge (MB)	OH	41.182″ N, 81.642″ W	Home	2009
Murray (MR)	IN	40.790″ N, 85.207″ W	Home	2009 and 2011
Pearsons (PS)	OH	41.643″ N, 83.435″ W	Home and Away	2009 and 2011
Reichert	IA	41.705″ N, 92.864″ W	Away	2011
TJ Dolan	ON	43.370″ N, 80.997″ W	Away	2009
Wabash Cannonball	OH	41.556″ N, 83.852″ W	Away	2009
Watson Road	ON	43.546″ N, 80.180″ W	Away	2009
Winchester (WN)	IN	41.172″ N, 81.202″ W	Home	2009
Yellowwood I (YW)	IN	39.219″ N, 86.342″ W	Home and Away	2009 and 2011
Yellowwood II	IN	39.165″ N, 86.343″ W	Away	2011
Yellowwood III	IN	39.218″ N, 86.340″ W	Away	2011
Yellowwood IV	IN	39.174″ N, 86.340″ W	Away	2011

dormancy differed between these experiments. In November 2009, we germinated seeds from 19 populations (Table 1). To break dormancy, we exposed seeds from one haphazardly chosen fruit per family (mean ± 1 SE families per population = 11.68 ± 1.01) to a 16:1:1 mixture of distilled water, bleach, and 70% ethanol, with 4 drops of triton X-100 detergent added per 2 L of solution (Dudle et al. 2001). In January 2011, we germinated seeds from 13 populations, including 6 of the populations from the first experiment (Table 1). To break dormancy, seeds from one fruit per family (mean families per population = 10.38 ± 0.76) were placed on moist filter paper in a petri dish, wrapped in parafilm, and stratified at 4 °C for eight weeks prior to planting (Johnston 1992). We sowed 100–200 seeds per maternal family into plug trays filled with Sunshine mix (Sun Gro Horticulture Canada Ltd., Vancouver, British Columbia, Canada). Trays were placed in standing water and their position on the greenhouse bench was randomized.

In December 2009 and February 2011, we transplanted 10–30 seedlings per family in 656 ml^3 Deepots (Stuewe and Sons Inc., Tangent, Oregon, USA) filled with Sunshine mix (n = 4263 seedlings in 2009, n = 2101 in 2011). Each plant was assigned a random position on the greenhouse bench. Plants were watered as necessary and fertilized with 8 pellets of Nutricote Total 13-13-13 (Plant Products, Brampton, Ontario, Canada). They were exposed to supplemental light (16 h days), and maintained at 23–25 °C during the day and 19–21 °C at night.

CROSSING DESIGN. During anthesis, we used hand-pollination to create within- and between-population crosses. Female plants from a subset of the populations in each experiment (hereafter Home populations; n = 9 in 2009 and n = 6 in 2011; Table 1) were used as pollen recipients for both within- and between-population crosses. Hermaphrodite plants from these Home populations were used as pollen donors for the within-population crosses. Hermaphrodites in the remaining populations (hereafter Away populations; n = 10 in 2009 and n = 7 in 2011; Table 1) were used as pollen donors for the between-population crosses. Four of the Home populations were also used as Away populations in some between-population crosses (Table 1).

To create the within-population crosses (mean per Home population = 7.22 ± 0.64), we hand pollinated four flowers on a female plant from a Home population with pollen from a hermaphrodite plant from the same population. We selected female and hermaphrodite plants from different maternal families to minimize inbreeding. All within-population crosses were made during the 2009 experiment. To create the between-population crosses (mean per Home population = 16.00 ± 2.31), we hand pollinated flowers on a female plant from a Home population with pollen from hermaphrodite plants from two different Away populations. Two flowers received pollen from each hermaphrodite, for a total of four flowers pollinated per female plant. Between-population crosses were made during both the 2009 and 2011 experiments. To control for variation in seed set along the raceme (Diggle 1995), whenever possible we pollinated four of the first ten flowers produced.

PERFORMANCE OF WITHIN- AND BETWEEN-POPULATION CROSSES. We tested for outbreeding depression by measuring four components of performance: seeds per fruit, seed size, germination success, and final aboveground biomass. We selected these components because they have been estimated in other studies of outbreeding depression (Hufford and Mazer 2003). In addition, seeds per fruit, seed size, germination success, and final aboveground biomass are all likely to affect the success of habitat restoration projects (Pywell et al. 2003, Menges 2008). Specifically, we assume that the best genotypes for habitat restoration projects are those that produce many, large seeds that have high germination success and accumulate a large aboveground biomass. Although there can be trade-offs between seed size and seed number that prevent plants from maximizing both performance components (*e.g.*, Paul-Victor and Turnbull 2009), the correlation between the number and size of seeds produced in our crosses was weak (Pearson product-moment correlation; $r = -0.199$, d.f. = 136, $P < 0.05$).

Because *L. siphilitica* seeds are small and numerous, we used digital imaging software to estimate seeds per fruit and seed size for all crosses. To estimate seeds per fruit, we used GeneSnap (Syngene, Frederick, Maryland, U.S.) to capture a digital image of the seeds from each fruit, and ProtoCOL (Synbiosis, Frederick, Maryland, U.S.) to count the image. To estimate seed size, we used OpenLab (PerkinElmer, Waltham, Massachusetts, U.S.) to capture an image of 20 haphazardly selected seeds per cross and estimate the length and width of each seed. We then used the formula for the area of an ellipse to calculate seed size. We estimated mean seeds per fruit and mean seed size for each cross.

We estimated germination success of seeds from a subset of the crosses created in the 2009 experiment. In June 2010, we germinated within- and between-population crosses from six of the Home populations (BVII, CAR, HR, MR, PS, and YW). Twenty stratified seeds from each cross were placed on moist filter paper in a petri dish. Each dish was assigned a random position on the greenhouse bench and watered as necessary. We counted the number of seeds that germinated in each dish every 2–3 d. The experiment was ended when no new seeds had germinated during the previous week. We estimated germination success as the proportion of seeds in each cross that germinated by the end of the trial.

We estimated final aboveground biomass of plants grown from the same subset of within- and between-population crosses used to estimate germination success. We chemically treated seeds from these crosses to break dormancy and sowed 100–200 seeds per maternal family into plug trays. In July 2010 we transplanted 20 seedlings from each family into Deepots (n = 2999 seedlings). Each plant was assigned a random position on the greenhouse bench and maintained as described in *Germination and Growth*, above. To estimate final biomass, we harvested the aboveground tissue of all surviving plants (n = 2187) after the topmost flower had opened or on December 14, 2010, whichever came first. Tissue was dried at 60 °C for 72 h and weighed. We calculated mean final aboveground biomass for each cross.

STATISTICAL ANALYSIS. In order to test whether geographically proximate populations were more ecologically similar than distant populations, we estimated the geographic and climatic distance between each pair of *L. siphilitica* populations used in the between-population crosses. Geographic distance was estimated as the straight-line distance between Home and Away populations. Climatic distance was estimated by extracting climate data for Home and Away populations from the WorldClim database (Hijmans et al. 2005). WorldClim contains estimates of average temperature and precipitation from 1950–2000. We then calculated the absolute value of the difference in annual mean temperature and annual precipitation between populations, resulting in two estimates of climatic distance. We focused on temperature and precipitation because they are associated with variation in *L. siphilitica*'s reproductive biology (Caruso and Case 2007). We estimated the Spearman rank correlation of geographic distance with temperature and precipitation to test whether geographically proximate *L. siphilitica* populations were more climatically similar to each other. We only analyzed the 129 unique pairs of *L. siphilitica* populations for which we made crosses. We restricted our analysis in this way because we were interested in the relationship between geographic distance, climatic distance, and performance, and we do not have estimates of performance for all possible crosses between the populations in Table 1.

We used linear mixed models to test whether crosses between geographically proximate and/or climatically similar populations had higher performance than crosses between distant and/or climatically dissimilar populations. We only included the between-population crosses in these analyses. We included Home population as a categorical fixed effect and the geographic or climatic distance between populations as a covariate, as well as their interaction. Because models that contained multiple distance estimates had excessive multicollinearity (variance inflation factors >10; Neter et al. 1990), we analyzed each distance estimate using a separate model. Maternal plant nested within Home population was included as a random effect. A significant distance or Home population $\times$ distance term indicates that the geographic or climatic distance between populations affects the performance of their offspring. When the distance or Home population $\times$ distance term was significant, we used the estimates of slope (β) from the linear model to determine whether performance increased or decreased with distance. Because the distribution of the germination success data was not suitable for analysis with parametric models, we present crossing distance analyses only for seeds per fruit, seed size, and final aboveground biomass.

Although seeds per fruit and seed size were measured in both the 2009 and 2011 experiments, we could not include year as a term in the linear mixed models. This is because we used a different subset of Home populations in the 2009 and 2011 between-population crosses, resulting in a correlation between year and Home population. To determine the effect of year on our results, we analyzed the 2009 crosses, the 2011 crosses, and the pooled 2009/2011 crosses. If the analyses of the pooled crosses were qualitatively similar to the analyses of the 2009 and 2011 crosses, then we present the results of the pooled analysis. If they were not similar, then we present the separate analyses of the 2009 and 2011 crosses.

We used two-way analysis of variance (ANOVA) to test whether seed number, seed size, and final aboveground biomass differed between within- and between-population crosses. We included cross type, Home population, and cross type $\times$ Home population as terms in the model. A significant cross type term indicates that performance differed between within- and between-population crosses. A significant cross type $\times$ Home population term indicates that the effect of cross type on performance differed among Home populations. When a cross type $\times$ Home population term was significant, we compared the performance of cross types within each home population using 1-d.f. contrasts. Because these contrasts were not orthogonal, we used the Bonferroni correction to control Type I error rates. Pairs of between-population crosses were nested within maternal plants and thus were not independent of each other. Consequently, prior to analysis we calculated mean seed number, seed size, and final aboveground biomass for each nested pair of between-population crosses.

Although seed number and seed size were measured in both the 2009 and 2011 experiments, we could not include year as a term in the ANOVA model. This is because the performance of all within-population crosses was measured in 2009, resulting in a perfect correlation between year and cross type. To determine the effect of year on our results, we analyzed the 2009 crosses and the pooled 2009/2011 crosses. If the analyses of the 2009 and 2009/2011 crosses were qualitatively similar, then we presented the pooled analysis. If they were not similar, then we presented the analyses of both the 2009 and 2009/2011 crosses.

Because the germination data were non-normal and heteroscedastic, we used a Wilcoxon test to determine if germination success differed between within- and between-population crosses. To avoid non-independence, we

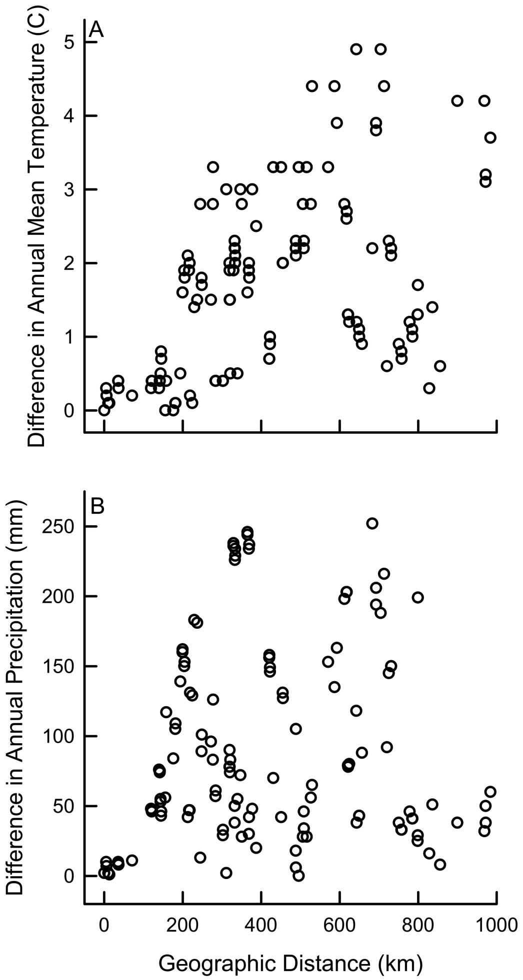

FIG. 1. Geographic and climatic distance between *Lobelia siphilitica* populations used in the between-population crosses. (A) Geographic distance versus the absolute value of the difference in annual mean temperature between populations. (B) Geographic distance versus the absolute value of the difference in annual precipitation between populations.

calculated mean germination success for each nested pair of between-population crosses prior to analysis. JMP version 10 (SAS, Cary, NC, U.S.) was used for all analyses.

Results. RELATIONSHIP BETWEEN GEOGRAPHIC AND CLIMATIC DISTANCE. The geographic distance between *L. siphilitica* populations was correlated with one of two measures of climatic distance. Geographically proximate populations had more similar annual mean temperature than distant populations (r_s = 0.529, d.f. = 127, $P < 0.001$; Fig. 1A). In contrast, geographically proximate populations did not have more similar annual precipitation than distant populations (r_s = 0.085, d.f. = 127, P = 0.332; Fig. 1B).

EFFECT OF GEOGRAPHIC DISTANCE ON PERFORMANCE. The geographic distance between *L. siphilitica* populations had a significant effect only on seeds per fruit (Table 2). Among all of the 2011 between-population crosses, crosses between distant populations set fewer seeds per fruit than crosses between geographically proximate populations (significant distance term). However, this effect of geographic distance on seeds per fruit varied among Home populations (significant Home population × distance term). In CAR, crosses with distant populations set significantly fewer seeds per fruit than crosses with geographically proximate populations. In contrast, in BVII, HR, and YW, crosses with distant populations set significantly more seeds per fruit than crosses with geographically proximate populations. Geographic distance also had an effect on the number of seeds per fruit among all of the 2009 between-population crosses; crosses between distant populations set more seeds per fruit than crosses between geographically proximate populations (significant distance term). However, this relationship did not remain significant after two influential data points were removed (results not shown). In contrast to seeds per fruit, seed size and final aboveground biomass were not affected by the geographic distance between populations.

EFFECT OF CLIMATIC DISTANCE ON PERFORMANCE. The difference in temperature between *L. siphilitica* populations had a significant effect on the number of seeds per fruit (Table 3). Among all of the 2011 between-population crosses, crosses between populations with dissimilar temperatures produced more seeds per fruit than crosses between populations with similar temperatures (significant distance term). However, this effect of temperature on seeds per fruit varied among Home populations (significant Home population × distance term). In CAR, crosses with populations with dissimilar temperatures produced more seeds per fruit than crosses with populations with similar temperatures, whereas in BVII, crosses with populations with dissimilar temperatures produced fewer seeds per fruit.

Table 2. Effect of the geographic distance between crossed populations on three components of performance in *Lobelia siphilitica*. To illustrate whether performance increased or decreased with distance, we included estimates of slope ($\beta \pm 1$ SE) from the linear model whenever the distance or Home population $\times$ distance term was significant.

Performance component	Source	d.f.$_{num}$, d.f.$_{demon}$	F	P
Seeds per fruit (2009 crosses)	Home population	8, 53.26	1.76	0.105
	Distance ($\beta = 1.15 \pm 0.51$)	1, 34.65	5.13	0.030
	Home population $\times$ Distance	8, 53.74	1.15	0.346
Seeds per fruit (2011 crosses)	Home population	5, 11.25	5.33	0.009
	Distance ($\beta = -0.42 \pm 0.14$)	1, 12.34	8.46	0.013
	Home population $\times$ Distance	5, 11.87	5.72	0.007
	$\beta_{BVII} = 0.64 \pm 0.27$*			
	$\beta_{CAR} = -2.39 \pm 0.54$***			
	$\beta_{HR} = 0.94 \pm 0.22$**			
	$\beta_{MR} = 0.09 \pm 0.26$			
	$\beta_{PS} = 0.15 \pm 0.28$			
	$\beta_{YW} = 0.57 \pm 0.22$*			
Seed size (2009 and 2011 crosses)	Home population	8, 74.78	3.15	0.004
	Distance	1, 60.63	0.11	0.740
	Home population $\times$ Distance	8, 106.50	0.58	0.793
Final aboveground biomass (2009 crosses)	Home population	5, 33.00	1.18	0.370
	Distance	1, 44.90	0.17	0.684
	Home population $\times$ Distance	5, 44.12	0.88	0.499

* $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$.

The difference in precipitation between *L. siphilitica* populations also had a significant effect on the number of seeds per fruit (Table 4). Among all of the 2009 between-population crosses, crosses between populations with dissimilar precipitation produced more seeds per fruit than crosses between populations with similar precipitation (significant distance term). However, this effect of precipitation on seeds per fruit varied among Home populations (significant Home population $\times$ distance term). In HR and KYR, crosses with populations with dissimilar precipitation set more seeds per fruit, whereas in MB crosses with populations with dissimilar precipitation set fewer seeds per fruit. In contrast, among all of the 2011 between-population crosses, there was no effect of precipitation on seeds per fruit (non-significant distance term), but the Home population $\times$ distance term was significant. In HR, crosses with populations with dissimilar precipitation set more seeds per fruit, whereas in CAR crosses with populations with dissimilar precipitation set fewer seeds per fruit.

The difference in temperature, but not precipitation, between *L. siphilitica* populations had a significant effect on seed size (Table 3, 4). Among all of the 2009 between-population crosses, crosses between populations with dissimilar temperatures produced smaller seeds than crosses between populations with similar temperatures (significant distance term; Table 3). However, this effect of temperature on seed size varied among Home populations (significant Home population $\times$ distance term; Table 3). In MB, crosses with populations with dissimilar temperatures produced smaller seeds than crosses with populations with similar temperatures, but the relationship between temperature and seed size was not significant within any other Home population (Table 3).

The difference in precipitation, but not temperature, between *L. siphilitica* populations had a significant effect on final aboveground biomass (Table 3, 4). Crosses between populations with dissimilar precipitation accumulated less biomass than crosses between populations that received similar amounts of precipitation (Table 4).

Performance of Within- versus Between-Population Crosses. One of four components of performance differed between within- and between-population crosses (Table 5). Among the 2009 crosses, between-population crosses produced larger seeds than within-population crosses (significant cross type term), but magnitude of this effect varied among Home populations (significant cross type $\times$ Home population term; Fig. 2B). In MR, between-population crosses produced 21.7% larger seeds than within-population crosses (1-d.f.

Table 3. Effect of climatic distance, estimated as the absolute value of the difference in annual mean temperature between crossed populations, on three components of performance in *Lobelia siphilitica*. To illustrate whether performance increased or decreased with distance, we included estimates of slope ($\beta \pm 1$ SE) from the linear model whenever the distance or Home population × distance term was significant.

Performance component	Source	df_{num}, df_{demon}	F	P
Seeds per fruit (2009 crosses)	Home population	8, 36.97	1.71	0.129
	Distance	1, 43.12	3.05	0.088
	Home population × Distance	8, 63.54	0.65	0.732
Seeds per fruit (2011 crosses)	Home population	5, 10.86	8.55	0.002
	Distance ($\beta = 159.37 \pm 40.40$)	1, 16.49	15.56	0.001
	Home population × Distance	5, 15.20	4.84	0.008
	$\beta_{BVII} = -240.44 \pm 76.97$**			
	$\beta_{CAR} = 327.49 \pm 91.72$**			
	$\beta_{HR} = 44.70 \pm 73.45$			
	$\beta_{MR} = 63.24 \pm 114.28$			
	$\beta_{PS} = -76.47 \pm 114.28$			
	$\beta_{YW} = -118.52 \pm 56.54$			
Seed size (2009 crosses)	Home population	8, 35.84	5.66	<0.001
	Distance	1, 41.13	0.04	0.838
	Home population × Distance	8, 64.46	2.32	0.030
	$\beta_{BVII} = 2.65 \times 10^{-3} \pm 7.14 \times 10^{-3}$			
	$\beta_{CAR} = 3.27 \times 10^{-4} \pm 6.72 \times 10^{-3}$			
	$\beta_{HR} = 1.19 \times 10^{-2} \pm 8.81 \times 10^{-3}$			
	$\beta_{KYR} = 1.70 \times 10^{-2} \pm 2.79 \times 10^{-2}$			
	$\beta_{MB} = -3.79 \times 10^{-2} \pm 9.96 \times 10^{-3}$***			
	$\beta_{MR} = -2.59 \times 10^{-3} \pm 5.83 \times 10^{-3}$			
	$\beta_{PS} = 8.17 \times 10^{-3} \pm 6.30 \times 10^{-3}$			
	$\beta_{WN} = 2.48 \times 10^{-3} \pm 6.72 \times 10^{-3}$			
	$\beta_{YW} = -1.95 \times 10^{-3} \pm 6.04 \times 10^{-3}$			
Seed size (2011 crosses)	Home population	5, 13.58	1.16	0.380
	Distance	1, 31.34	0.39	0.537
	Home population × Distance	5, 28.21	0.59	0.710
Final aboveground biomass (2009 crosses)	Home population	5, 28.39	1.69	0.168
	Distance	1, 47.27	0.30	0.584
	Home population × Distance	4, 44.38	1.05	0.399

* $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$.

contrast, $F_{1,95} = 9.54$, $P = 0.003$), but cross type did not affect seed size in the other Home populations. In contrast to seed size, the effect of cross type on germination success (Wilcoxon test; $Z = 1.48$, $n = 71$, $P = 0.140$; Fig. 2D), seeds per fruit (Fig. 2A), and final aboveground biomass (Fig. 2E) was small and non-significant; between-population crosses had 0.18% higher germination and produced 2.5% more seeds/fruit than within-population crosses, whereas within-population crosses had 2.4% higher final biomass than between-population crosses.

Discussion. We found mixed support for the assumption that geographically proximate populations are more ecologically similar than distant populations. Nearby *L. siphilitica* populations had more similar annual mean temperatures than distant populations, but there was no relationship between geographic distance and the difference in annual precipitation. In addition, except for seeds per fruit, the geographic distance between populations did not affect the performance of between-population crosses. These results are consistent with past studies, which have found that the geographic distance between populations is (1) an imperfect proxy for their ecological similarity (*e.g.*, Montalvo and Ellstrand 2000, Raabova et al. 2007), and (2) a relatively poor predictor of the performance of translocated seed sources (*e.g.*, Montalvo and Ellstrand 2000, Bischoff et al. 2006, Raabova et al. 2007, Lawrence and Kaye 2011). Consequently, if the goal is to use seeds from populations that are ecologically similar to the transplant site, then our results suggest that *L. siphilitica* seeds from geographically distant but climatically similar populations could be suitable for use in restoration projects.

We also found mixed support for the assumption that offspring of crosses between

Table 4. Effect of climatic distance, estimated as the absolute value of the difference in annual precipitation between crossed populations, on three components of performance in *Lobelia siphilitica*. To illustrate whether performance increased or decreased with distance, we included estimates of slope ($\beta \pm 1$ SE) from the linear model whenever the distance or Home population $\times$ distance term was significant.

Performance component	Source	df_{num}, df_{demon}	*F*	*P*
Seeds per fruit (2009 crosses)	Home population	8, 50.19	1.18	0.332
	Distance ($\beta = 1.95 \pm 0.61$)	1, 67.70	10.18	0.002
	Home population $\times$ Distance	8, 49.86	2.74	0.014
	$\beta_{BVII} = -1.72 \pm 1.08$			
	$\beta_{CAR} = -0.69 \pm 3.18$			
	$\beta_{HR} = 3.91 \pm 1.55$*			
	$\beta_{KYR} = 4.38 \pm 1.83$*			
	$\beta_{MB} = -2.78 \pm 1.08$*			
	$\beta_{MR} = 0.60 \pm 2.20$			
	$\beta_{PS} = -1.07 \pm 0.97$			
	$\beta_{WN} = -0.89 \pm 1.52$			
	$\beta_{YW} = -1.73 \pm 0.87$			
Seeds per fruit (2011 crosses)	Home population	5, 8.36	5.73	0.014
	Distance	1, 11.87	0.40	0.541
	Home population $\times$ Distance	5, 10.88	3.60	0.036
	$\beta_{BVII} = -0.17 \pm 0.68$			
	$\beta_{CAR} = -4.49 \pm 1.16$**			
	$\beta_{HR} = 2.50 \pm 1.02$*			
	$\beta_{MR} = 0.71 \pm 1.10$			
	$\beta_{PS} = 0.83 \pm 0.67$			
	$\beta_{YW} = 0.62 \pm 0.68$			
Seed size (2009 and 2011 crosses)	Home population	8, 65.04	3.20	0.004
	Distance	1, 103.00	0.63	0.429
	Home population $\times$ Distance	8, 105.50	1.01	0.436
Final aboveground biomass (2009 crosses)	Home population	5, 38.82	2.12	0.084
	Distance ($\beta = -6.96 \times 10^{-3} \pm 3.41 \times 10^{-3}$)	1, 48.82	4.15	0.047
	Home population $\times$ Distance	5, 43.97	1.59	0.182

* $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$.

individuals from ecologically similar populations have higher fitness than offspring of crosses between ecologically dissimilar populations. Crosses between climatically dissimilar *L. siphilitica* populations produced smaller seeds and accumulated less final biomass than crosses from climatically similar populations. These results are consistent with the hypothesis that mating between individuals from different populations introduces maladaptive alleles into locally adapted populations, resulting in outbreeding depression (Hufford and Mazer 2003). However, outbreeding depression often was detected in only a subset of the Home *L. siphilitica* populations. In other Home populations, crosses between climatically dissimilar populations had higher performance (*i.e.* heterosis), or there was no relationship

Table 5. Effects of cross type (within- vs. between-population) on three components of performance in *Lobelia siphilitica*. Error d.f. = 119 for seeds per fruit, 119 for seed size (2009 and 2011 crosses), 95 for seed size (2009 crosses), and 68 for final aboveground biomass.

Performance component	Source	d.f.	SS	*F*	*P*
Seeds per fruit (2009 and 2011 crosses)	Cross type	1	6.67×10^{3}	0.14	0.713
	Home population	8	1.24×10^{6}	3.14	0.003
	Cross type $\times$ Home population	8	2.93×10^{5}	0.74	0.653
Seed size (2009 and 2011 crosses)	Cross type	1	2.61×10^{-3}	3.60	0.060
	Home population	8	2.18×10^{-2}	3.76	<0.001
	Cross type $\times$ Home population	8	9.93×10^{-3}	1.71	0.103
Seed size (2009 crosses)	Cross type	1	3.72×10^{-3}	5.13	0.026
	Home population	8	2.43×10^{-2}	4.18	<0.001
	Cross type $\times$ Home population	8	1.25×10^{-2}	2.15	0.039
Final aboveground biomass (2009 crosses)	Cross type	1	7.63×10^{-2}	0.16	0.692
	Home population	5	16.30	6.76	<0.001
	Cross type $\times$ Home population	5	1.43	0.59	0.706

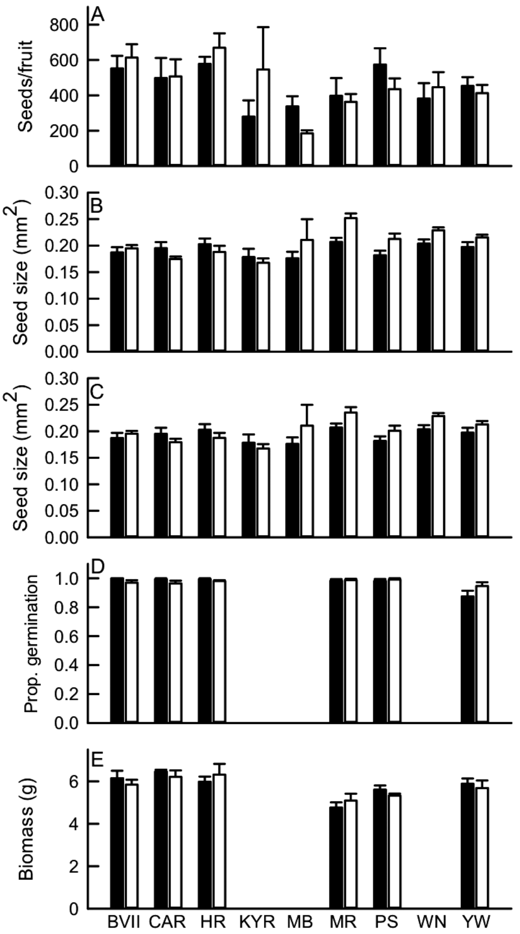

FIG. 2. Performance of crosses within (solid bars) and between (open bars) *Lobelia siphilitica* populations. (A) Seeds per fruit (2009 and 2011 crosses). (B) Seed size (2009 crosses). (C) Seed size (2009 and 2011 crosses). (D) Proportion germination (2009 crosses). (E) Final aboveground biomass (2009 crosses). Population abbreviations follow Table 1.

between climatic distance and performance. These results are consistent with recent studies (*e.g.*, Willi and Fischer 2005, Willi et al. 2007, Pickup and Young 2008, Pickup et al. 2013) showing that the effect of crossing distance on fitness varies among populations. Consequently, while mating with climatically dissimilar populations does not decrease the performance of all *L. siphilitica* populations, our results do suggest that the potential for outbreeding depression needs to be considered when selecting seed sources for restoring *L. siphilitica* populations.

In contrast to our analysis of the between-population crosses, we did not find any evidence for outbreeding depression when we compared the performance of within- vs. between-population crosses. Between-population crosses did not have lower performance than within-population crosses. Instead, crossing between *L. siphilitica* populations increased one of four components of performance (*i.e.*, heterosis). Although comparing within- vs. between-population crosses is a common approach to testing for outbreeding depression (*e.g.*, Byers 1998, Raabova et al. 2009), this approach can confound inbreeding depression within populations with outbreeding depression between populations (Montalvo and Ellstrand 2001). In addition, analyses of the performance of different between-population crosses are more informative for choosing seed sources for ecological restoration projects (Montalvo and Ellstrand 2001).

Our inferences about outbreeding depression in *L. siphilitica* are limited in two ways. First, we cannot exclude the possibility that we would have found additional evidence for outbreeding depression if we had grown our within- and between-population crosses in a field environment (*e.g.*, Fenster and Galloway 2000, Montalvo and Ellstrand 2001). This is because the expression of outbreeding depression caused by gene flow between locally adapted populations may vary with the growth environment (Hufford and Mazer 2003). Second, we cannot exclude the possibility that we would have found additional evidence for outbreeding depression in second-generation (*i.e.*, F_2) or later between-population crosses (*e.g.*, Parker 1992, Fenster and Galloway 2000). This is because any outbreeding depression caused by the disruption of co-adapted gene complexes may not be expressed until interacting genes recombine (Lynch 1991). Consequently, our results represent a conservative estimate of the extent of outbreeding depression in *L. siphilitica*.

In conclusion, we found mixed support for the recommendation that habitat restoration projects use seed collected from geographically proximate populations. Geographic distance was an imperfect proxy for climatic distance in *L. siphilitica*, and crosses with climatically dissimilar populations decreased the performance of *L. siphilitica* from some but not all Home populations. These results have two implications for the development of guidelines for selecting seed sources for restoration projects. First, our results support the recommendation (*e.g.*, Bischoff et al. 2006, Broadhurst et al. 2008) that factors other than geographic distance need to be considered when selecting seed sources for restoration projects, because geographically proximate populations may not always be the most climatically similar. Second, our results support the recommendation (*e.g.*, Raabova et al. 2007, Vander Mijnsbrugge et al. 2010) that seeds for restoration projects come from populations that are ecologically similar to the transplant site, because ecological similarity can be a good predictor of performance. More generally, our study highlights how measuring performance in common garden experiments can be helpful in selecting appropriate seed sources for restoration projects (*e.g.*, Gordon and Rice 1998, Bischoff et al. 2006, Gibson et al. 2013).

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